Nonlocal Band-Weighted Iterative Spectral Mixture Model for Hyperspectral Imagery Denoising

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***Abstract*— Although efficient hyperspectral image (HSI) denoising relies on complete and accurate description and modeling the spatial–spectral signal in HSI, the current approaches do not fully account for key characteristics of HSI, i.e., the mixed spectra effect, the spatial nonstationarity effect, and noise variance heterogeneity effect. To address this issue, this article presents a linear spectral mixture model with nonlocal means constraint (LSMM-NLMC), with the following advantages. First, LSMM-NLMC can effectively learn the signal in mixed pixels in HSI by estimating clean endmembers and abundances for image restoration. Second, LSMM-NLMC can efficiently address nonstationary spatial correlation effect by imposing NLMC on the latent scene signal. Last, LSMM-NLMC provides accurate noise characterization by accounting for noise variance heterogeneity effect using a band-dependent noise model and a band-weighted Mahalanobis distance for similarity measurement. A novel optimization method based on the expectation–maximization (EM) algorithm and the purified means approach is used to efficiently solve the resulting maximum a posterior (MAP) problem. The experiments on both simulated and real HSI data sets demonstrate that the visual quality and denoising accuracy are significantly improved by the proposed LSMM-NLMC compared with previous methods.**

***Index Terms*— Band-dependent noise, hyperspectral image (HSI) denoising, linear spectral mixture model (LSMM), mixed pixel, spatial nonstationarity, the Mahalanobis distance.**

I. Introduction

HYPERSPECTRAL image (HSI), which has both spatial information and spectral information of scenery, provides high spatial–spectral sensitivity to different materials [1], [2] and has been widely used in various tasks [3]–[7]. Nevertheless, one critical hampering factor is the noise effect in HSI, which causes the visual quality degradation of HSI and large ambiguities and uncertainties in HSI applications. As such, denoising is a crucial task for effective HSI applications.

In past decades, many research studies have been conducted for HSI denoising [8]–[11]. Zheng *et al.* [12] first proposed that the HSI denoising problem is transformed into a band fusion problem . Given that noise may occur only in a certain band in real circumstance, information bands with superior quality are selected to construct a nonlinear regularized energy minimization function for estimating the target noised band. But the determination of the target noised band is a key problem. Liu and Lee [13] proposed a 3D atrous denoising convolution neural network (3DADCNN) that simultaneously extracts the more different scale context spectral and spatial information of HSI by enlarging the receptive field without significantly increasing the number of network parameters. Yuan *et al.* [14] first proposed a spectral–spatial kernel-based method for HSI denoising to maintain the spectral correlations in spectral dimension using the spectral mean estimation and to match the original structure between two spatial dimensions using self-similarity of spectra. Most methods differentiate signal and noise from a statistical perspective [15]. As a typical example, principal component analysis (PCA) addresses statistically uncorrelated signal and assumes that the signal has a larger variance than noise [16]. In recent studies, PCA has been treated as a base model for HSI denoising and many efficient HSI denoising methods have been proposed by integrating PCA into wavelet approaches, which addresses the localized signal and assumes signal lying in lower frequency domain than noise. For example, PCA and 2-D wavelet are combined in simultaneous dimensionality reduction and HSI denoising [17], [18]; the 2-D bivariate wavelet thresholding method is used to remove the noise in lower-energy PCA output channels [19]; the hybrid spatial–spectral method is proposed to link both spatial and spectral domains, where the PCA and bivariate wavelet shrinkage method are performed in the 2-D spatial domain, and 2-D PCA transformation is used in the 1-D spectral domain [20]. However, the statistical characteristics of noise and signal cannot be easily modeled and discriminated in the complex heterogeneous spatial–spectral HSI data, and more efficient HSI denoising approaches can possibly be derived by exploiting the intrinsic characteristics of HSI.

The mixed spectra effect, the spatial nonstationarity effect, and noise variance heterogeneity effect in HSI are critical, and

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useful characteristics that can be leveraged to improve HSI denoising, which, however, have not been fully investigated in previous studies. First, an efficient HSI denoising approach should effectively learn the signal in mixed pixels in HSI by estimating clean endmembers and abundances for clean image reconstruction. Due to the ground complexity with various materials and target types and the insufficient spatial resolution of a hyperspectral imaging spectrometer, there are many mixed pixels in HSI. As a result, the signal in the spectral domain is essentially heterogeneous with significant spatial–spectral variability, and it is difficult for HSI denoising algorithms to model and capture such heterogeneity and variability. Instead of relying on statistical characteristics, efficient HSI denoising should take advantages of the underlying physical characteristics of the mixed pixels by investigating the spectral mixture model (SMM) that describes the observed mixed spectrum as a function of endmembers (i.e., the spectra of constituent ground types or pure materials) weighted by their factional abundances [21] with the noise. Then, the noise-free estimation of the heterogeneous mixed spectrum can be achieved by reconstruction using the endmembers and abundances. Therefore, the investigation on the use of SMM for accurately and efficiently capturing the spectral heterogeneity and variations in spectrally mixed signal is critical.

Second, an efficient HSI denoising approach should be able to capture the nonstationary spatial signals for improving signal preservation and noise characterization. An HSI usually includes many different land cover types and various targets, which are especially true in urban areas, leading to various edges, boundaries as well as textural and structural patterns where abrupt intensity changes can be observed. These nonstationary spatial signals [22], [23] poses significant challenges on HSI denoising algorithms. Comparing with the local stationary approaches, which assume that spatially close pixels have a high spatial correlation effect, the nonlocal approaches calculate the degree of spatial correlation effect based on not only the spatial-closeness but also the spectral similarity between two pixels in a large searching window. As a result, the nonlocal methods can be better capture the heterogeneous or nonstationary spatial signals in HSI whose spatial correlation effect is not isotropic and distance-dependent but varies with directions and spatial locations [14], [24], [25]. Therefore, how to integrate the nonlocal approach into the SMM framework for improved HSI denoising is an important research issue.

Third, an efficient HSI denoising approach should have accurate noise characterization to be used as model regularization and constraints that effectively isolate the noise from signal. If noise is insufficiently characterized, it will be mistreated as being signal and preserved, leading to weak noise removal. In contrast, overcharacterization of noise leads to overestimation of noise and bad signal preservation. Due to the varying scattering mechanism of the different spectral bands and the existence of the junk bands in HSI [26], the noise levels of different spectral bands in HSI are not the same, which is called here the noise variance heterogeneous issue. Therefore, designing a statistical model that is able to accurately accommodate and characterize this noise variance heterogeneity is important. Moreover, the accurate statistical noise model also benefits the defining of the metric for measuring the spectral similarity between two pixels in HSI. Instead of using the Euclidean distance that assumes isotropic diagonal covariance matrix [27], [28], the Mahalanobis distance that use band-dependent noise variances to adjust the importance of each spectral band for calculating the spectral similarity is more preferable.

Recently, Xue *et al.* [29] considered both nonlocal similarity across the spatial domain and global correlation among the spectral domain into hierarchical least square-based nonnegative tucker decomposition method. Bai *et al.* [30] proposed a nonlocal low-rank regularized CANDECOMP/PARAFAC tensor decomposition (NLR-CPTD) denoising model, which utilizes both nonlocal similarity across the spatial space and global redundancy along the spectral space, and was solved by a hierarchical alternative least square-based nonnegative tucker decomposition to achieve better denoising effect. Different from the optimization models and the solving targets of these methods, this article presents a linear SMM with nonlocal means constraint (LSMM-NLMC) method based on the above considerations for HSI denoising. The proposed approach can achieve accurate modeling of the complex heterogeneous spatial–spectral signal in HSI for efficient HSI denoising, by jointly accounting for three key characteristics of HSI, i.e., the mixed spectra effect, the spatial nonstationarity effect, and the noise variance heterogeneity effect. The experiments on both simulated and real HSI data sets demonstrate that the proposed LSMM-NLMC can outperform the other state-of-the-art HSI denoising algorithms. The main contributions of this article are summarized as follows.

1) An iterative approach based on LSMM is designed for retrieving the endmembers and the abundances maps in HSI to effectively account for the heterogeneity and variability in mixed pixels in HSI for enhanced HSI denoising.
2) An nonlocal means constraint (NLMC) is designed to serve as a regularization on the latent scene signal, such that the nonstationary signals such as the edges and structural patterns can be better restored and preserved in the denoised result.
3) A band-dependent noise model and a band-weighted similarity measure are designed to better characterize the heterogeneous noise variance in HSI.
4) A novel optimization method based on the expectation-maximization (EM) algorithm and the purified means approach is used to efficiently solve the resulting maximum a posterior (MAP) problem.

This article is organized as follows. The MAP framework of HSI denoising is described in Section II. Section III introduces the LSMM-NLMC method for HSI denoising, and the optimization of this method is presented in Section IV. Section V presents experimental setting and results. Section VI concludes the method of this article.

II. MAP Framework for HSI Denoising

The HSI denoising aims at recovering noise-free HSI $\boldsymbol{X}$ from observed noisy HSI $\boldsymbol{Y}$ based on the following genera-

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tive model:

$$\boldsymbol{Y} = \boldsymbol{X} + \boldsymbol{n} \tag{1}$$

where $\boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{n} \in \mathbb{R}^{P\times N}$, N and P are, respectively, the number of pixels and the number of bands in HSI, with $N = m1 \times m2$, and $m1$ and $m2$ being, respectively, the number of columns and the number of rows in HSI. The noise satisfies zero-mean Gaussian distribution with variance $\boldsymbol{\Lambda}$, i.e., $\boldsymbol{n} \sim \mathcal{N}(0, \boldsymbol{\Lambda})$, and thereby the data likelihood has the following form:

$$\begin{aligned} p(\boldsymbol{Y}|\boldsymbol{X}) &= \frac{1}{\sqrt{(2\pi)^{pn}|\boldsymbol{\Lambda}^N|}} e^{-\frac{1}{2}\mathrm{tr}((\boldsymbol{Y}-\boldsymbol{X})^T\boldsymbol{\Lambda}^{-1}(\boldsymbol{Y}-\boldsymbol{X}))} \\ &= \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi^P|\boldsymbol{\Lambda}|}} e^{-\frac{1}{2}(\boldsymbol{Y}_i-\boldsymbol{X}_i)^T\boldsymbol{\Lambda}^{-1}(\boldsymbol{Y}_i-\boldsymbol{X}_i)} \end{aligned} \tag{2}$$

where $\mathrm{tr}(\cdot)$ is the trace operation, and Y_i denotes the site of the ith pixel in HSI that is a $P \times 1$ vector.

According to MAP, $\hat{\boldsymbol{X}}$ can be estimated by

$$\hat{\boldsymbol{X}} = \arg\max_{\boldsymbol{X}} p(\boldsymbol{Y}|\boldsymbol{X})p(\boldsymbol{X}) \tag{3}$$

where $p(\boldsymbol{X})$ is the prior distribution of the noise-free HSI. Equation (3) is equivalent to

$$\hat{\boldsymbol{X}} = \arg\min_{\boldsymbol{X}} \sum_{i=1}^{N} (\boldsymbol{Y}_i - \boldsymbol{X}_i)^T\boldsymbol{\Lambda}^{-1}(\boldsymbol{Y}_i - \boldsymbol{X}_i) + \lambda R(\boldsymbol{X}) \tag{4}$$

where $R(\boldsymbol{X}) = -log(p(\boldsymbol{X}))$ is the regularization and constraints on $\boldsymbol{X}$, and λ is the weight of the regularization item.

To efficiently and accurately separate $\boldsymbol{X}$ from $\boldsymbol{n}$, $R(\boldsymbol{X})$ in (4) and $\boldsymbol{\Lambda}$ in (2) have to reflect the key characteristics of HSI (i.e., the mixed spectra effect, the spatial nonstationarity effect, and the noise variance heterogeneity effect), which is achieved in Section III by the proposed approach.

III. LSMM-NLMC for HSI Denoising

A. LSMM for Spectral Heterogeneity and Variations

To effectively learn the signal in mixed pixels in HSI, in the proposed HSI denoising approach, the spectral heterogeneity and variations in spectrally mixed pixels is addressed by adopting the LSMM to describe $\boldsymbol{X}$ in (1) and (4), that is

$$\boldsymbol{X} = \boldsymbol{A}\boldsymbol{S}, \ \text{s.t. } \boldsymbol{A}, \boldsymbol{S} \geq 0 \tag{5}$$

where $\boldsymbol{A} = [\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_K]$, with K being the number of endmembers and $\boldsymbol{a}_k$ being the kth endmember that is a $P \times 1$ vector, and $\boldsymbol{S} \in \mathbb{R}^{K\times N}$ is the abundance matrix. To achieve physical meanings, we impose nonnegative constraint (NC) on endmembers and abundance, i.e., $\boldsymbol{A}, \boldsymbol{S} \geq 0$. For the ith pixel in HSI, we have

$$\boldsymbol{Y}_i = \boldsymbol{A}\boldsymbol{s}_i + \boldsymbol{n}_i, \ \text{s.t. } \boldsymbol{A}, \boldsymbol{s}_i \geq 0 \tag{6}$$

where $\boldsymbol{Y}_i$ and $\boldsymbol{n}_i$ represent, respectively, a $P \times 1$-dimensional measured spectrum vector and noises at site i; $\boldsymbol{s}_i$ represents $K \times 1$ dimensional abundance vector whose jth element s_{ij} is associated with $\boldsymbol{a}_j$, and $\mathbf{1}^T\boldsymbol{s}_i = 1$, where the mean of the proportion of all the endmembers in the ith mixed spectrum is one.

We therefore, update (4) with the following formulation:

$$\begin{aligned} \hat{\boldsymbol{X}} = \arg\min_{\boldsymbol{A},\boldsymbol{S}} &\sum_{i=1}^{N} (\boldsymbol{Y}_i - \boldsymbol{A}\boldsymbol{s}_i)^T\boldsymbol{\Lambda}^{-1}(\boldsymbol{Y}_i - \boldsymbol{A}\boldsymbol{s}_i) \\ &\text{s.t. } \boldsymbol{X} = \boldsymbol{A}\boldsymbol{S}, \quad \boldsymbol{A}, \boldsymbol{S} \geq 0. \end{aligned} \tag{7}$$

The regulation $R(\boldsymbol{X})$ in (4) is implemented in (7) by $\boldsymbol{A}$, $\boldsymbol{s}_i \geq 0$ which takes advantages of the underlying physical characteristics of the mixed pixels for better addressing capturing the spectral heterogeneity and variations in spectrally mixed signal.

B. NLMC for Spatial Nonstationarity

To capture the nonstationary spatial signals for improving signal preservation and noise characterization, the proposed approach adopts an NLMC to improve the LSMM regulation in (5) and (7), that is

$$E(\boldsymbol{s}_i|\boldsymbol{S}) = \frac{1}{M} \sum_{j\in\varphi(i)} w_{i,j}\boldsymbol{s}_j \tag{8}$$

where $E(\boldsymbol{s}_i|\boldsymbol{S})$ represents the conditional expectation of the ith abundance $\boldsymbol{s}_i$, given abundances at all the other pixels in HSI; $\varphi(i)$ is the collection of all neighboring pixels of the ith pixel; and $w_{i,j}$ is the weight of the jth neighbor $\boldsymbol{s}_j$, which is implemented as a probabilistic similarity between the ith pixel and the jth pixel.

As we can see, (8) essentially requires that the conditional expectation of $\boldsymbol{s}_i$ equals the weighted average of all its neighboring pixels. To capture the nonstationary spatial signals in HSI, there are two important issues, i.e., (A) how do we calculate the weight $w_{i,j}$, and (B) how do we define the neighborhood $\varphi(i)$.

1) Spatial–Spectral Probabilistic Similarity Measure: We designed the following probabilistic similarity measure to implement the weight $w_{i,j}$ in (8):

$$w_{i,j} = e^{-D_{i,j}/h} \tag{9}$$

where $D_{i,j}$ denotes the spectral dissimilarity between the ith pixel and the jth pixel, and h is a scaling constant.

The spectral dissimilarity $D_{i,j}$ can be implemented as the distance between the ith pixel and the jth pixel in spectral domain, that is

$$D_{i,j}^{\mathrm{spe}} = \sqrt{(\boldsymbol{Y}_i - \boldsymbol{Y}_j)^T\boldsymbol{\Lambda}^{-1}(\boldsymbol{Y}_i - \boldsymbol{Y}_j)} \tag{10}$$

However, (10) only measures the spectral dissimilarity, and does not consider the influence of the local spatial textures. So, we design the following spatial–spectral probabilistic similarity likelihood, that is:

$$D_{i,j}^{\mathrm{ss}} = \sum_{f_0=1}^{f^2} \sqrt{(\boldsymbol{Y}_{i+f_0} - \boldsymbol{Y}_{j+f_0})^T\boldsymbol{\Lambda}^{-1}(\boldsymbol{Y}_{i+f_0} - \boldsymbol{Y}_{j+f_0})} \tag{11}$$

where f_0 is used to iterate through all local square window (with the width being f) centered at the ith and the jth pixels. Therefore, $D_{i,j}^{\mathrm{ss}}$ measures the spatial–spectral dissimilarity between the ith 3-D cube centered at the ith pixel and the jth 3-D cube centered at the jth pixel. Therefore, $w_{i,j}$ defined

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using $D_{i,j}^{\mathrm{ss}}$ can ensure that if two pixels are similar, then they not only have similar spectral characteristics but also have similar local texture patterns.

Note that both $D_{i,j}^{\mathrm{spe}}$ and $D_{i,j}^{\mathrm{ss}}$ are Mahalanobis distances which can better account for the noise influence on distance measures than the Euclidean distance that is commonly adopted in the literature [28], [31].

2) Nonlocal Neighborhood: The neighborhood $\varphi(i)$ in (8) essentially defines the range of the spatial correlation effect, and it also defines which pixels will be used for calculating the expectation of $\boldsymbol{X}_i$. Instead of using the local means approach, we use the nonlocal approach which can better handle the spatial nonstationarity effect. First, in terms of range, spatial correlation effect is much larger, instead of using a very small local window, the nonlocal approach considers a much larger window with size $l \times l$ for finding the neighbors. Second, in terms of determining the neighbors, instead of using all pixels in the $l \times l$ as neighbors, the nonlocal approach just selects the first M pixels that have the highest spatial–spectral probabilistic similarities defined in Section III-B1.

Based on (8), we can update (7) with the following formulation:

$$\begin{aligned}\hat{\boldsymbol{X}} = \arg\min_{\boldsymbol{A},\boldsymbol{S},\boldsymbol{\Lambda}} & \sum_{i=1}^{N} (\boldsymbol{Y}_i - \boldsymbol{A}\boldsymbol{s}_i)^T \boldsymbol{\Lambda}^{-1} (\boldsymbol{Y}_i - \boldsymbol{A}\boldsymbol{s}_i) \\ & + \lambda \left(\boldsymbol{s}_i - \frac{1}{M} \sum_{j \in \varphi(i)} w_{i,j} \boldsymbol{s}_j \right)^T \boldsymbol{\Omega} \left(\boldsymbol{s}_i - \frac{1}{M} \sum_{j \in \varphi(i)} w_{i,j} \boldsymbol{s}_j \right) \\ & \text{s.t. } \boldsymbol{X} = \boldsymbol{A}\boldsymbol{S}, \quad \boldsymbol{A}, \boldsymbol{S} \geq 0 \end{aligned} \tag{12}$$

where the second term in (12) is used to impose the NLMC in (9) in the objective function, and $\boldsymbol{\Omega} = P\boldsymbol{A}^{\boldsymbol{T}}\boldsymbol{\Lambda}\boldsymbol{A}/\mathrm{tr}(\boldsymbol{A}^{\boldsymbol{T}}\boldsymbol{\Lambda}\boldsymbol{A})$ is a diagonal matrix that defines the degrees of smoothness-constraint for different bands.

Therefore, the regulation $R(\boldsymbol{X})$ in (4) is implemented in (12) by both the NCs and the NLMCs to address both the spectral heterogeneity in mixed signal and the spatial nonstationary correlation effect.

C. Band-Dependent Noise Variance

To address the noise variance heterogeneous for accurate noise characterization and effective similarity measure in (11), we implement $\boldsymbol{\Lambda}$ in as a nonisotropic diagonal matrix

$$\boldsymbol{\Lambda} = \begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_P^2 \end{pmatrix} \tag{13}$$

where σ_p^2 is the noise variance at the pth band. The use of different noise variances $\{\sigma_p^2 | p = 1, 2, \ldots, P\}$ for different bands allows a better accommodation of the noise variance heterogeneity effect in HSI and thereby prevents the noise overestimation in some bands and the noise underestimation in some other bands, as the case that will happen when using the same noise variance for all bands.

Moreover, using $\boldsymbol{\Lambda}$ in (13) leads to the band-weighted Mahalanobis distances in (12)

$$\begin{aligned} & (\boldsymbol{Y}_i - \boldsymbol{A}\boldsymbol{s}_i)^T \boldsymbol{\Lambda}^{-1} (\boldsymbol{Y}_i - \boldsymbol{A}\boldsymbol{s}_i) \\ & = \|\boldsymbol{\Lambda}^{-1/2} (\boldsymbol{Y}_i - \boldsymbol{A}\boldsymbol{s}_i)\|_2^2 \end{aligned} \tag{14}$$

$$\begin{aligned} & \times \left(\boldsymbol{s}_i - \frac{1}{M} \sum_{j \in \varphi(i)} w_{i,j} \boldsymbol{s}_j \right)^T \boldsymbol{\Omega} \left(\boldsymbol{s}_i - \frac{1}{M} \sum_{j \in \varphi(i)} w_{i,j} \boldsymbol{s}_j \right) \\ & = \left\| \boldsymbol{\Omega}^{1/2} \left(\boldsymbol{s}_i - \frac{1}{M} \sum_{j \in \varphi(i)} w_{i,j} \boldsymbol{s}_j \right) \right\|_2^2 \end{aligned} \tag{15}$$

where $||\cdot||_2^2$ means L2 norm. Equations (14) and (15) adjust the weights of each bands in the data likelihood and the nonlocal constraint according to the band-dependent noise variances $\{\sigma_p^2 | p = 1, 2, \ldots, P\}$.

IV. Model Optimization

A. EM Algorithm Optimization Procedure

Given the definitions in Sections III-A–III-C, the MAP in (4) can be reformulated as

$$\begin{aligned} \hat{\boldsymbol{X}} = \arg\min_{\boldsymbol{A},\boldsymbol{S},\boldsymbol{\Lambda}} & \sum_{i=1}^{N} \|\boldsymbol{\Lambda}^{-1/2} (\boldsymbol{Y}_i - \boldsymbol{A}\boldsymbol{s}_i)\|_2^2 \\ & + \lambda \left\| \boldsymbol{\Omega}^{1/2} \left(\boldsymbol{s}_i - \frac{1}{M} \sum_{j \in \varphi(i)} w_{i,j} \boldsymbol{s}_j \right) \right\|_2^2 \\ & \text{s.t. } \boldsymbol{X} = \boldsymbol{A}\boldsymbol{S}, \quad \boldsymbol{A}, \boldsymbol{S} \geq 0 \end{aligned} \tag{16}$$

where the unknown parameters $\Theta = \{\boldsymbol{A}, \boldsymbol{S}, \boldsymbol{\Lambda}\}$, and the number of which is more than the number of observations $\boldsymbol{Y}$. So, the estimation of Θ based on $\boldsymbol{Y}$ is an ill-posed optimization problem, which can be solved by the EM algorithm by iterating the E-step as described in Section IV-C and M-step in Section IV-D.

B. Initialization

Before EM iterations, $\boldsymbol{A}$ and $\boldsymbol{\Lambda}$ need to be initialized. A recommended initialization approach is to randomly select M spectra from HSI as initial $\boldsymbol{A}$ and use the intensity variances of pixels in a homogeneous area in HSI as initial estimation of the band-dependent noise variances in $\boldsymbol{\Lambda}$.

C. E-Step

In E-step, $\boldsymbol{s}_i$ is estimated given $\boldsymbol{A}$ and $\boldsymbol{\Lambda}$ by minimizing the following objective function:

$$\begin{aligned} \hat{\boldsymbol{s}}_i = \arg\min_{\boldsymbol{s}_i} & \|\boldsymbol{\Lambda}^{-1/2} (\boldsymbol{Y}_i - \boldsymbol{A}\boldsymbol{s}_i)\|_2^2 \\ & + \lambda \left\| \boldsymbol{\Omega}^{1/2} \left(\boldsymbol{s}_i - \frac{1}{M} \sum_{j \in \varphi(i)} w_{i,j} \boldsymbol{s}_j \right) \right\|_2^2, \quad \text{s.t.} \quad \boldsymbol{s}_i \geq 0 \end{aligned} \tag{17}$$

which can be solved by addressing a constrained least square estimation problem. Note that $\{\boldsymbol{s}_i\}$ and $\{\boldsymbol{s}_j | j \in \varphi(i)\}$ are simultaneously estimated. However, we only keep $\{\boldsymbol{s}_i\}$. The update of $\boldsymbol{S}$ is achieved by updating $\{\boldsymbol{s}_i | i = 1, 2, \ldots, N\}$ sequentially in this step. In this step, before estimating $\boldsymbol{s}_i$, the estimation of $w_{i,j}$ and $\varphi(i)$ is defined in Section II-B.

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D. M-Step

In M-step, $\boldsymbol{A}$ and $\boldsymbol{\Lambda}$ are updated based on $\boldsymbol{S}$. First, we estimate $\boldsymbol{\Lambda}$ as

$$\hat{\boldsymbol{\Lambda}} = \text{diag}(\text{var}(\boldsymbol{Y} - \boldsymbol{AS})) \tag{18}$$

where var$(\cdot)$ returns the variances of all bands, and diag(B) returns a diagonal matrix with diagonal elements being B.

We estimate $\boldsymbol{A}$ using a purified means approach [32], which estimates an endmember $\boldsymbol{a}_k$ as the weighted mean value of the "purified" pixels $\{\boldsymbol{b}_i^k | i = 1, 2, \ldots, N\}$ associated with the kth endmember

$$\hat{\boldsymbol{a}}_k = \frac{\sum_{i=1}^{N} \boldsymbol{b}_i^k s_{ik}}{\sum_{i=1}^{N} s_{ik}} \tag{19}$$

where $\boldsymbol{b}_i^k$ is the ith purified pixel that is only due to the kth endmember by removing the spectral contribution from all the other endmembers

$$\boldsymbol{b}_i^k = \boldsymbol{Y}_i - \sum_{t \neq k}^{K} \boldsymbol{a}_t s_{it}. \tag{20}$$

Note that the elements in $\boldsymbol{A}$, i.e., $\{\boldsymbol{a}_k | k = 1, 2, \ldots, K\}$, are updated sequentially in an element-wise manner. The update of $\hat{\boldsymbol{a}}_k$ relies on the values of $\{\boldsymbol{a}_{t \neq k}\}$. Once $\hat{\boldsymbol{a}}_k$ is updated, its updated value is used to update the other endmembers.

E. Algorithm Summary

The proposed denoising algorithm is achieved by repeating the E-step and M-step defined above to solve (16), which is summarized in Algorithm 1.

Algorithm 1 EM Algorithm

Input: $\boldsymbol{Y}$, λ, M, h, f, l;
Output: $\boldsymbol{S}$, $\boldsymbol{A}$;
Initialization of $\boldsymbol{A}$ and $\boldsymbol{\Lambda}$
while $i \leq maxiter$ **do**
  E-step:
  for i=1:N **do**
    obtain $\left\{ w_{i,j} \middle| j \in \varphi(i) \right\}$ using Eq. 9
    estimate $\boldsymbol{s_i}$
  end for
  M-step:
  update $\boldsymbol{\Lambda}$ using Eq. 18;
  for $k = 1, 2, \ldots, K$ **do**
    update $\boldsymbol{a}_k$ using Eq. 20 and Eq. 19;
  end for
end while

V. Experiments

In this section, we illustrate the experimental design and discuss the experimental results.

A. Experimental Design

The proposed LSMM-NLMC algorithm is compared with several other advanced HSI denoising approaches on various HSI data sets, i.e., simulated HSI, airborne HSI scene, and terrestrial HSI scene, and the results are evaluated regarding both visualization and numerical measures.

1) Methods Selection: To evaluate the performance of the LSMM-NLMC method, the obtained denoising results of the LSMM-NLMC method is compared with the results of the most used method, such as the LSMM with local spatial constraint (LSMM-LSC) [33], the local mean method [34], the nonlocal mean (NLM) method [24], the local spatial–spectral PCA method (SSPCA) [16], parallel factor analysis (PARAFAC) [35], tensor dictionary learning (TDL) [36], and structured low-rank matrix analysis (SLRMA) method [37] as comparable methods and through both quantitative assessment and visual assessment. The LSMM-LSC method is an integration of the LSMM and LSC to explain the influence of NLMC in the LSMM for denoising. The local mean method and the NLM method are based on the filter, and in this article, they are compared with each other based on their denoising results to evaluate the effectiveness of nonlocal neighborhood in HSI denoising. The SSPCA method adopts the local neighborhood spectral as a constraint for denoising to test the LSMM performance in solving the mixture spectrum. Here, the SSPCA method is realized using the mature toolkit in MATLAB R2016a. The SSPCA and LSMM-LSC methods local neighborhood spectral set include four neighborhoods of each spectrum. In the LSMM-NLMC method, the $maxiter$ and h are set to be 20 and 10, respectively. The PARAFAC model reduces additive white Gaussian Noise from HSI by exploiting its decomposition uniqueness and its single rank character. The TDL method combinatorically considers two intrinsic characteristics underlying an HSI: the nonlocal similarity over space and the global correlation across the spectrum. The SLRMA method fully uses local and nonlocal similarities together with structures inherently contained in the low-rank matrix in regularization term of it, obtained the denoised HSI adaptively using an efficient optimization method.

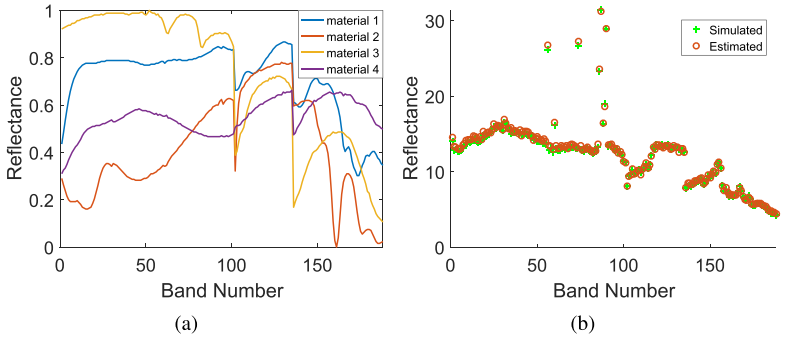
2) Numerical Measures: Three parameters are verified using the simulated data sets. The mean peak signal to noise ratio (MPSNR), mean structural similarity (MSSIM), and the mean spectral angle distance (MSAD), which denote the average PSNR, SSIM, and SAD values in all the bands, are used to evaluate the denoising performance by the quantitative assessment of the simulation results. The three evaluation indicators are, respectively, calculated by

$$\begin{aligned} \text{MPSNR} &= \frac{1}{P} \sum_{p=1}^{P} \text{PSNR}_p \\ \text{MSSIM} &= \frac{1}{P} \sum_{p=1}^{P} \text{SSIM}_p \\ \text{MSAD} &= \frac{1}{N} \sum_{i=1}^{N} \text{SAD}_i \end{aligned} \tag{21}$$

where PSNR_p and SSIM_p are the PSNR and SSIM values at the pth band, respectively. SAD_i denotes the spectral angle distance of the ith spectrum. And the larger the SAD is, the worse the denoising result is.

3) Data Descriptions: In this article, the simulated data set and two types of the real data set, including the Indian Pines

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(a) (b)

Fig. 1. (a) Four spectra of the simulated data set randomly selected from the USGS spectral library. (b) Simulated and estimated noise variance in different bands on the simulated data set experiment at snr = 20.



(a) (b) (c)

Fig. 2. Spectral bands of test data sets in real data set experiments. (a) 120th band of Indian Pines data set. (b) 60th band of SOC710-VP data set 1. (c) 60th band of SOC710-VP data set 2.

scene [38] and portable hyperspectral camera SOC710-VP scene, are denoised by the LSMM-NLMC method and several above-listed denoising methods.

a) Simulated data set: The simulated data set is a mixture of four spectral signatures provided by the U.S. geological survey (USGS) digital spectral library with different proportion [33], [39], and it is utilized in parameter sensitivity analysis and experiment on simulation data. It consists of 188 bands, some junk bands are removed, and the size of each band is 58 × 58 pixels. The four spectral signatures containing 188 bands are shown in Fig. 1(b).

The simulated data set with five noise levels is used to test the stability of the LSMM-NLMC method. For different bands, the noise intensity is unequal. The additional simulated noise is generated by the function, $\boldsymbol{n} = \text{randn}(0, \boldsymbol{\Lambda})$, where randn generates the random numbers, and $\boldsymbol{\Lambda}$ is the simulated noise variance $\boldsymbol{\Lambda} = \text{diag}(E[\boldsymbol{X}^T\boldsymbol{X}]/(N \times 10^{(}\boldsymbol{c}/10)))$. $\boldsymbol{c} = \sin(T) + \text{snr}$, where $T = \text{rand}[0, 2\pi)$, and snr[5, 10, 20, 30, 40]. snr determines the SNR values of the simulated data set and indicates the noise level of the simulated data set. The lower the snr value is, the higher the noise level is. Further, randn is used to generate nonmean Gaussian random numbers with variance $\boldsymbol{\Lambda}$; rand randomly selects P numbers from the range $[0, 2\pi)$. The simulated noise variance and estimated noise variance are shown in Fig. 1(a).

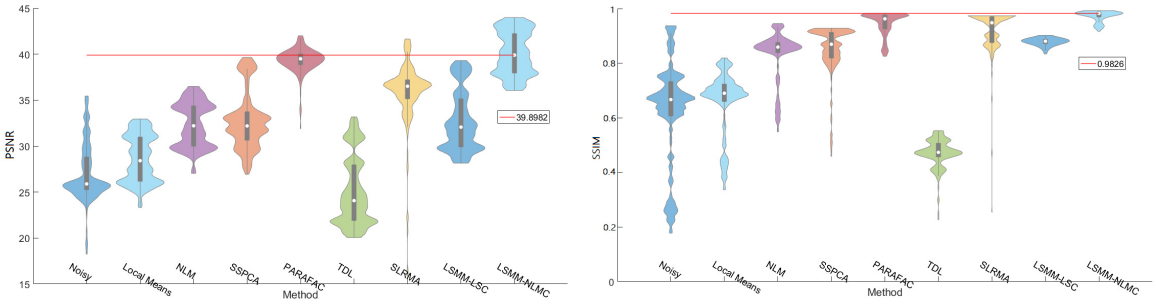
b) Indian pines scene: The Indian Pines data set is collected by the airborne visible and infrared imaging spectrometer instrument (AVIRIS) over the Indian Pines test site in Northwestern Indiana, USA, in 1992, at a spatial resolution of 20 m, covering each band with 145 × 145 pixels, and including 220 spectral bands in the wavelength range 400–2500 nm. The scene is shown in Fig. 2(b).

c) Portable hyperspectral camera SOC710-VP scene: The portable hyperspectral camera SOC710-VP data set consisted of two data sets, which includes 128 spectral bands with 4.6875-nm width and wavelengths from 400 to 1000 nm and are made by ourselves using the original HSI provided by the Bejing AZUP scientific limited. SOC710-VP data set 1 mainly includes minerals and vegetation, and the size of each band is 131 × 201 pixels. SOC710-VP data set 2 mainly includes artificial building and vegetation; the size of each band is 121 × 131 pixels. Their scenes are illustrated in Fig. 2(b) and (c), respectively.

B. Experiment on Simulated Data Set

To comprehensively compare the denoising effect of the proposed method and a few other HSI denoising methods, such as the quantitative assessments, spatial visual, and spectral difference, are adopted to evaluate the HSI quality before and after denoising. The quantitative assessment indexes, such as MPSNR (PSNR), MSSIM (SSIM), and MSAD (SAD), are used to reveal the noise removal ability of denoising methods. Namely, MSSIM and SSIM indicate the capacity of preserving the image structure information. The structure information and noise intensity from spectral bands of HSI are presented by the visualization of denoising results. The spectral difference is determined by computing MSAD (SAD) and contrasting the spectral curves of the truth and denoised data sets.

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(a) (b)

Fig. 3. PSNR (a) and the SSIM (b) obtained by different methods, in different bands of the simulated data set at snr = 20. For both measures, the lines corresponding to the LSMM-NLMC method are above those of the other methods, indicating that the LSMM-NLMC method outperforms the other methods in terms of both noise removal and detail preservation.

TABLE I

ASSESSING INDEXES OF DIFFERENT DENOISING METHODS ON SIMULATED DATA SETS AT DIFFERENT NOISE LEVELS

Signal-to-Noise level	Assessing indexes	Local means	NLM	SSPCA	PARAFAC	TDL	SLRMA	LSMM-LSC	LSMM-NLMC
snr=5	MPSNR	19.6005	22.6081	29.8586	24.6452	24.5067	**31.0038**	29.1905	29.0187
	MSSIM	0.2987	0.4382	**0.8097**	0.4941	0.383	0.7892	0.7801	0.7704
	MSAD	9.4689	6.6469	2.5857	7.2526	5.5813	**2.4249**	2.8175	2.9072
snr=10	MPSNR	23.5527	26.7719	31.2785	29.5259	23.8595	33.4989	31.1119	**33.3298**
	MSSIM	0.466	0.6501	0.8632	0.7209	0.3728	0.876	0.8672	**0.9081**
	MSAD	5.877	4.0448	2.1933	4.0653	5.8159	1.9981	2.0691	**1.8264**
snr=20	MPSNR	27.9037	31.6897	31.824	39.3392	25.0357	35.9128	31.8545	**40.0021**
	MSSIM	0.7039	0.8554	0.8869	0.9436	0.4703	0.9179	0.9041	**0.9829**
	MSAD	3.2989	2.1869	2.0663	0.9426	4.8356	2.2143	1.8315	**0.7325**
snr=30	MPSNR	29.8049	33.8342	31.869	**43.5292**	25.4214	36.8921	32.0805	42.8599
	MSSIM	0.8249	0.9304	0.8895	0.9713	0.5089	0.9314	0.9134	**0.9929**
	MSAD	2.4202	1.5784	2.0576	0.5579	4.6124	2.2992	1.79	**0.4952**
snr=40	MPSNR	30.1649	34.1178	31.8761	45.0165	25.4871	36.0067	32.9463	**46.9707**
	MSSIM	0.8504	0.9417	0.8897	0.9766	0.5165	0.9182	0.9376	**0.9968**
	MSAD	2.2833	1.5028	2.0562	0.4707	4.5808	2.4023	1.5254	**0.2649**

Table I provides the MPSNR, MSSIM, and MSAD values of the denoised data set obtained by the eight denoising methods for five noise intensity simulated data sets. As shown in Table I, it can be noticed that the proposed LSMM-NLMC method obtains the highest MPSNR (MSSIM) and lowest MSAD values in most cases. The local mean method and the NLM method generate the worse index values (lower MPSNR and MSSIM and higher MSAD) than the proposed method. This is because these single-band methods operate on each band independently and neglect the band relationship in HSIs. The SSPCA, PARAFAC, and LSMM-LSC achieve better index values (higher MPSNR and MSSIM, and lower MSAD) than the former two methods and worse index values (lower MPSNR and MSSIM, and higher MSAD) than the proposed method. This is because these methods treat the HSI as a whole for considering the band relationship but neglect nonlocal spatial information, leading to the obscure edges between different targets. The TDL method obtains the worse index values (lower MPSNR and MSSIM, higher MSAD) than the proposed LSMM-NLMC method for ignoring the different noise intensities of different bands in HSI. Due to lacking of accurate representation for the sparsity of the low-rank matrix, the SLRMA method generates worse index values (lower MPSNR and MSSIM, and higher MSAD) than the proposed method for simulated HSI with different noise levels, except snr = 5. The proposed LSMM-NLSC method can exploit the nonlocal spatial correlation and adaptively adjust the noise level for the different bands. It is clear that the proposed LSMM-NLSC method can achieve a better denoising performance than other competing methods in the relatively stable and better index values (higher MPSNR and MSSIM, lower MSAD) when dealing with different noise levels.

Furthermore, Fig. 3 shows the violin plots of PSNR and SSIM values for all spectral bands and SAD values for all spectral curves at snr = 20, where the horizontal axis represents the method and the vertical axis represents the PSNR/SSIM/SAD values. Those values are calculated using the original clean HSI and the denoised HSI by different methods. The violin plot is actually a combination of a box plot and the nuclear density plot, where the box plot shows the

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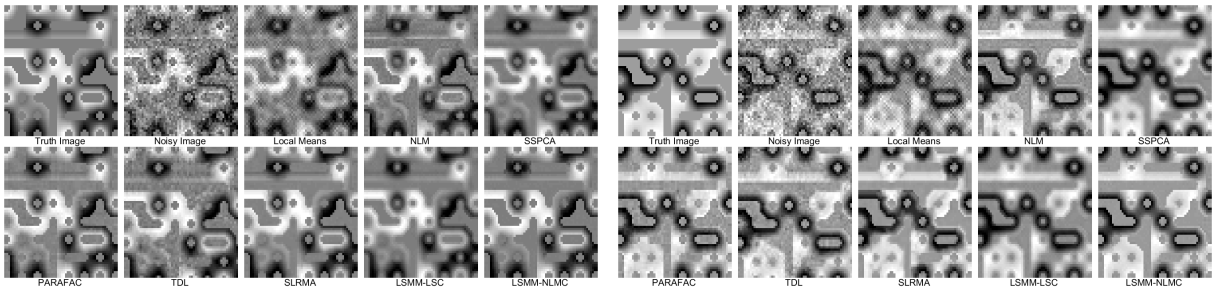


Fig. 4. Band images of the simulated data set at snr = 20 before and after denoising using the eight methods. (a) 51st band. (b) 140th band.

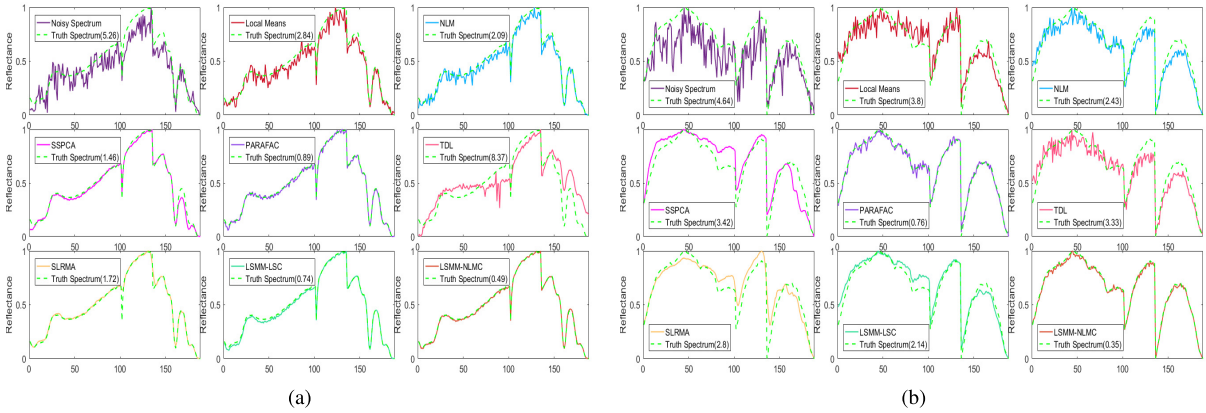


Fig. 5. Spectral signatures of the simulated data set at snr = 20 at different sites before and after denoising using the eight methods. The values in the brackets is SAD of the spectral. (a) Site at (28, 45). (b) Site at (41, 7).

position of the quantile, and the violin plot is used to display the data distribution and its probability density. The location with higher density can be easily found in the violin plot. In Fig. 3, the white point is the median value. The gray box is the range from the lower quartile to the upper quartile, and the thin gray line extending from it represents the 95% confidence interval. The outer shape is the kernel density estimation. The red line represents the best median value of PSNR, SSIM, and SAD for clearly comparing the performance of different methods. As shown in Fig. 3, the proposed LSMM-NLSC method has the highest PSNR and SSIM median values, and the lowest SAD median values among all methods in the simulated HSI at snr = 20. Besides, the distributions of the PSNR and SSIM values are more concentrate and larger than other denoising methods, and the distribution of the SAD is concentrate and lower than these comparison methods. It means that, compared with other methods, the proposed LSMM-NLSC method performs stably on the three measures for the simulated data set with noise level snr = 20.

To further clarify the superiority of the proposed LSMM-NLSC method, two spectral bands (i.e., the 51st and 140th bands) are randomly selected from original clean simulated HSI, noisy HSI, and denoised HSI with different methods as visual results and presented in Fig. 4. In Fig. 4, it can be seen that the LSMM-NLMC method approximates the original noiseless spectral band images compared with the other methods, which means that it can simultaneously suppress the noise and preserve the details of the original HSI. However, the local mean method does not perform well in both denoising and detail preservation. Although the NLM, PARAFAC, and TDL methods remain clear boundaries, they fail in noise removal. On the contrary, the SSPCA and LSMM-LSC methods perform well in HSI denoising but blur the edges of HSI. The newly proposed SLRMA method has a better visual result than the other method used for comparison, but it is still a loser than the proposed method in detail preservation.

To demonstrate the superiority of the proposed LSMM-NLSC method on avoiding spectral distortion, two spectral curves at different sites are randomly selected from original clean simulated HSI, noisy HSI, and denoised HSI with different methods as spectral difference results and presented in Fig. 5. The spectral reflectance values are shown on the vertical axis, and the band numbers on the horizontal axis. The SAD values are also shown in the legend. The smaller the SAD value is, the smaller the difference result is, and the better the spectral recovery result is. In Fig. 5, it can be seen that the spectral curve of the LSMM-NLMC method denoising is smoother and closer to the real spectral curve than those of the other methods, and its SAD value is the lowest. Fig. 5 further verifies the effectiveness of the proposed LSMM-NLMC method in avoiding spectral distortion.

With a decrease in the noise intensity, the denoising accuracy and effectiveness of the proposed LSMM-NLMC

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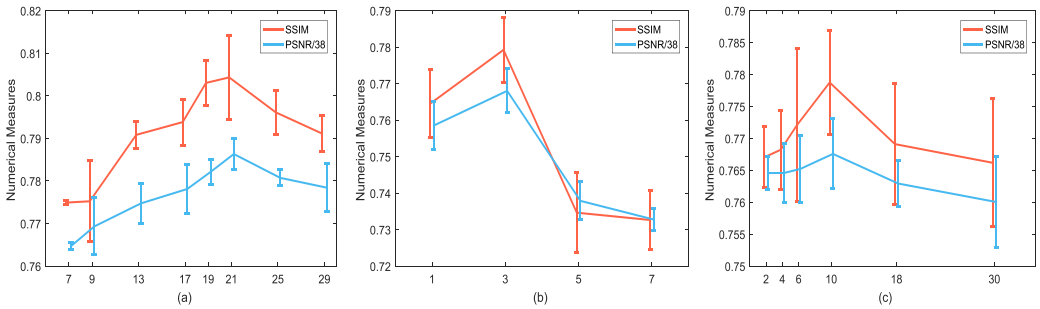


Fig. 6. Numerical measurement error of the proposed method at the Mahalanobis distance. (a) Dependence of the PSNR and SSIM on the "search window" size on the simulated data, when all the other parameters are fixed at the default values. (b) Numerical measurement error as a function of the 3-D-block size when all the other parameters are fixed at the default values. (c) Numerical measurement error as a function of the number of neighborhood spectra when all the other parameters are fixed at the default values. The center of the bar represents the mean value of numerical results for all the bands, while the height represents the standard deviation.

method are much higher than those of the other denoising methods. By using the proposed method, the noise is effectively removed, and more high-frequency information of an image is reserved in the visual results. The spectral signature of the original HSI is more effectively reconstructed by the LSMM-NLMC than by the other methods. Namely, the LSMM-NLMC not only reduces spectral noise but also mitigates spectral distortion.

C. Parameters Sensitivity Analysis

In this section, the influence of parameters, i.e., neighbor spectral number M, 3-D-block size f, and "search window" size l, on the simulated data set at snr $=$ 5 is evaluated using two indicators, PSNR and SSIM. The influence of the Mahalanobis distance and the Euclidean distance on the LSMM-NLMC denoising effectiveness is also analyzed.

The experimental results of the proposed method on the simulated data set at snr $=$ 5, including the influence of the neighbor spectral number M, 3-D-block size f, and "search window" size l on the LSMM-NLMC performance at $M = 2$, 4, 6, 10, 18, 30, $f = 1$, 3, 5, 7, and $l = 7$, 9, 13, 17, 19, 21, 25, 29, are presented in Fig. 6. With an increase in the parameter values, the LSMM-NLMC performance first increases and then decreases. As shown in Fig. 6(a), the best result is at $l = 21$, where MPSNR $=$ 29.72 and MSSIM $=$ 0.80. More reliability similar spectral can be obtained from a large "search window". However, an over large "search window" can increase the computation cost and bring more dissimilar spectral than similar spectral. As shown in Fig. 6(b), the best performance is at $f = 3$ where MPSNR $=$ 29.17 and MSSIM $=$ 0.78. Smaller f indicates weaker spatial constraints, and over larger f chooses some dissimilar spectra to weaken spatial constraint, so the suitable neighborhood number is very important. Lastly, as shown in Fig. 6(c) the best MPSNR and MSSIM of 29.18 and 0.78, respectively, are achieved when $M = 10$. Lower M indicates that there is not enough NLMC for the LSMM, and higher M may cause some dissimilar spectral in the neighborhood spectral set. Therefore, $l = 21$, $f = 3$, and $M = 10$ are used as optimal parameters in the following experiments.

To enhance the LSMM-NLMC performance, the Euclidean distance is replaced by the Mahalanobis distance. Namely,

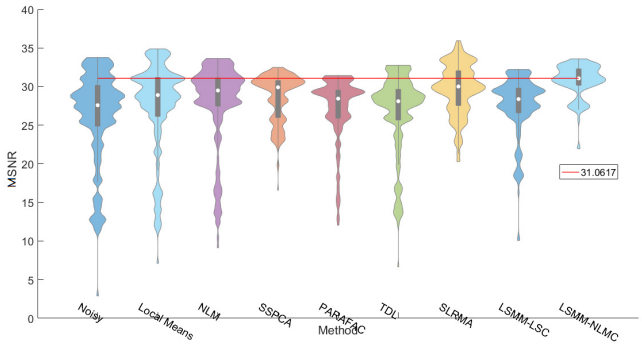


Fig. 7. SNR (dB) obtained by different methods, in different bands of the Indian pines data.

TABLE II
COMPARISON OF THE LSMM-NLMC METHOD FOR THE MAHALANOBIS DISTANCE AND THE EUCLIDEAN DISTANCE

Distance		MPSNR	MSSIM
Mahalanobis distance	Mean	**29.36**	**0.7870**
	Standard deviation	0.12	0.0072
Euclidean distance	Mean	28.14	0.7232
	Standard deviation	0.23	0.0119

MPSNR and MSSIM of LSMM-NLMC are calculated for a noise-free simulated data set for both distances, and the obtained results are given in Table II. The denoising accuracy comparison shows that denoising is more accurate for the Mahalanobis distance than that for the Euclidean distance. In the case of the Mahalanobis distance, the noise variance heterogeneity effect in the LSMM-NLMC denoising is efficiently adjusted.

D. Experiment on AVIRIS Indian Pines Data Set

The performance of the LSMM-NLMC method is further investigated on the Indian Pines data set where each band involves different noise levels. Due to the lack of reference data set of the real HSI data set, the mean of SNR (MSNR) of ten randomly selected homogeneous regions is chosen as a measure for a quantitative assessment of each band. The spectral band image visualization result is utilized to

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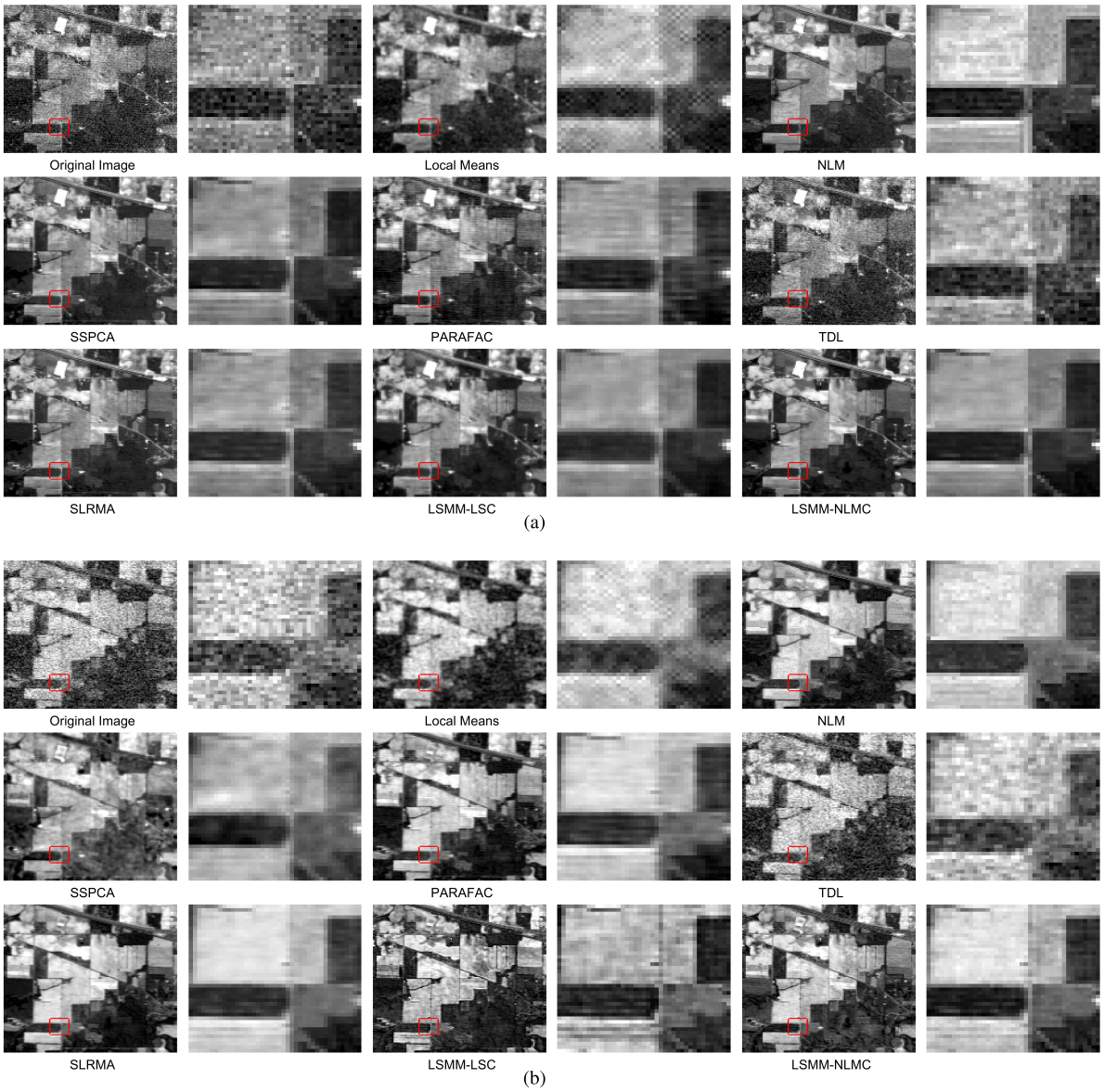


Fig. 8. Spectral band images of the Indian Pines before and after denoising by the eight methods. (a) 3rd band. (b) 110th band.

demonstrate the structural information preservation and noise removal of the denoising method.

Fig. 7 shows the violin plot of MSNR values for all the spectral bands of the Indian Pines data set after and before the denoising. We can see that the proposed LSMM-NLMC method obtains the maximum MSNR median value, the most concentrated distribution, and the shortest interquartile range among other competing methods, which means that it does denoise well in all bands of the Indian Pines data set. The proposed LSMM-NLMC method achieves success on adaptively denoising for the different bands and preserving detail information. Obviously, the proposed LSMM-NLMC method is competitive in noise removal for real HSI.

Fig. 8 displays visual results of two randomly selected spectral bands, i.e., the 3rd and 110th bands. It can be observed that the local means and TDL methods generate the worst denoising results and the SSPCA method loses the most information. The spectral band images of these methods still contain a lot of residual noises and blurred edges between different ground targets. The PARAFAC method does not achieve well denoising for each band and brings artifacts in some band images (i.e., the 3rd band). The LSMM-LSC method removes noise but somewhat blurs the spatial structural information. The NLM and SLRMA methods do not remove the noise in HSI completely. The results indicate that the LSMM-NLMC method obtains a much better visual result in noise reduction and structure information preservation than these corrival denoising methods. Thus, the LSMM-NLMC method is a suitable approach to denoise real HSI with a large variety of different noise level bands.

Compared with the other denoising methods, the proposed LSMM-NLMC method provides denoising results with higher visual quality and more distinct and sharper edges. Even in the bands with severe noise, the LSMM-NLMC still

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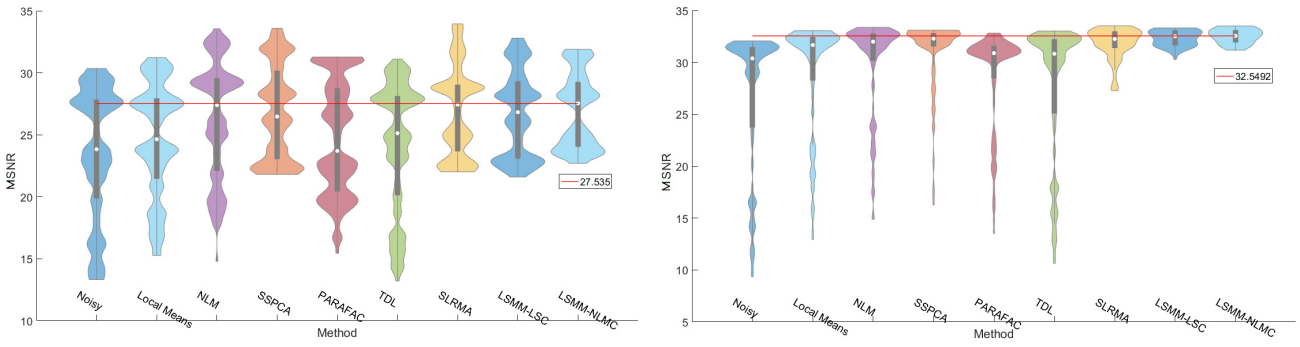


Fig. 9. SNR obtained by different methods in different bands. (a) SOC710-VP data set 1. (b) SOC710-VP data set 2.

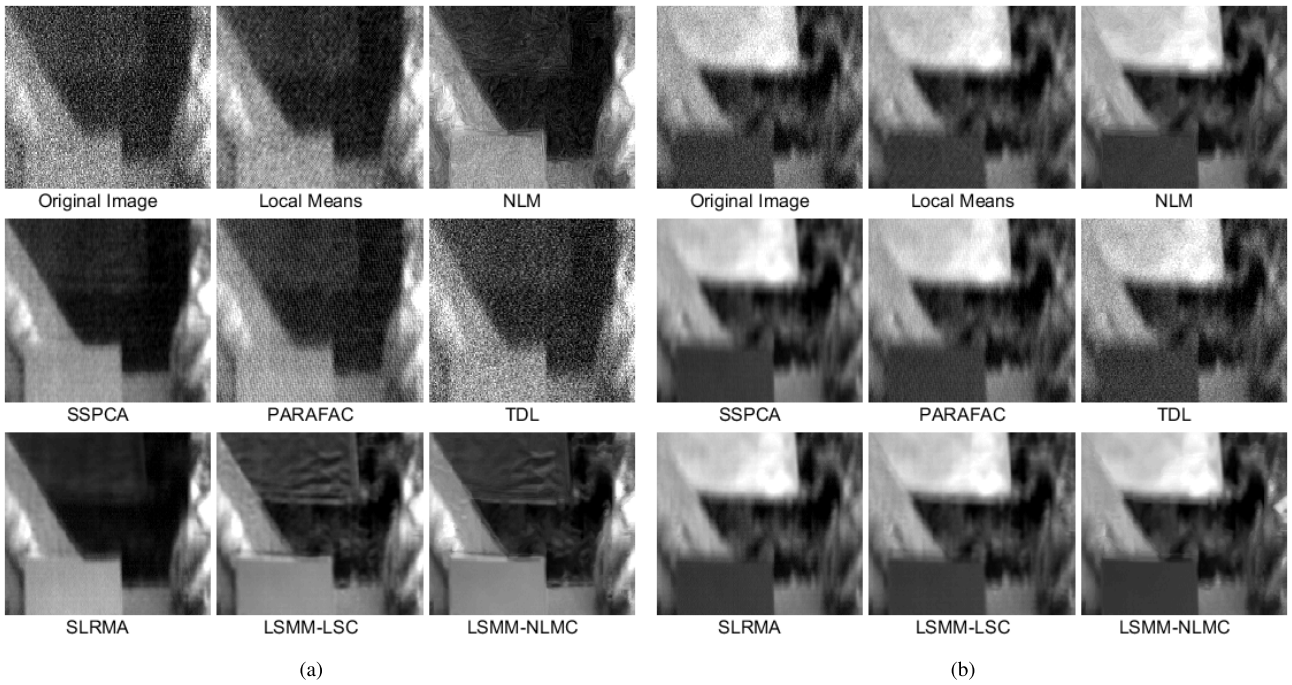


Fig. 10. Spectral band images of the SOC710-VP data set 1 before and after denoising with the eight methods. (a) 4th band. (b) 119th band.

can effectively restore the details. Thus, the LSMM-NLMC method is a suitable approach to denoise HSI with a large variety of different noise level bands.

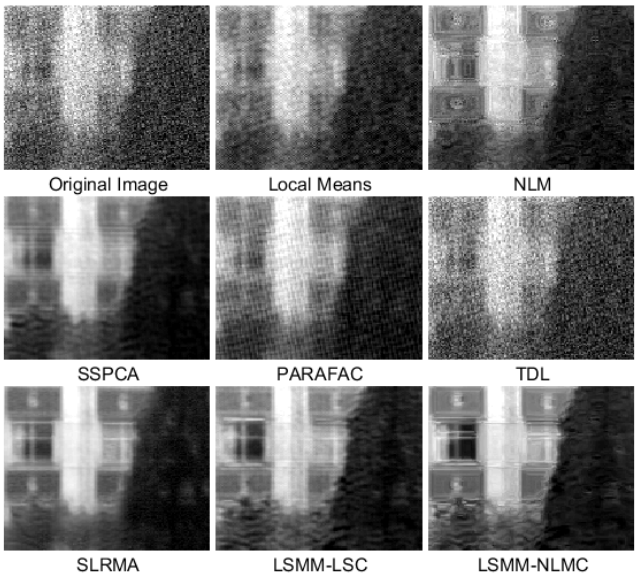
E. Experiments on Terrestrial SOC710-VP HSI Data Set

The SOC710-VP data set, including data set 1 and data set 2 and containing two different scenes, is used to further test the denoising performance of the LSMM-NLMC on portable hyperspectral camera data set. The MSNR value and spectral band image visualization result are used to assess the denoising ability of different methods.

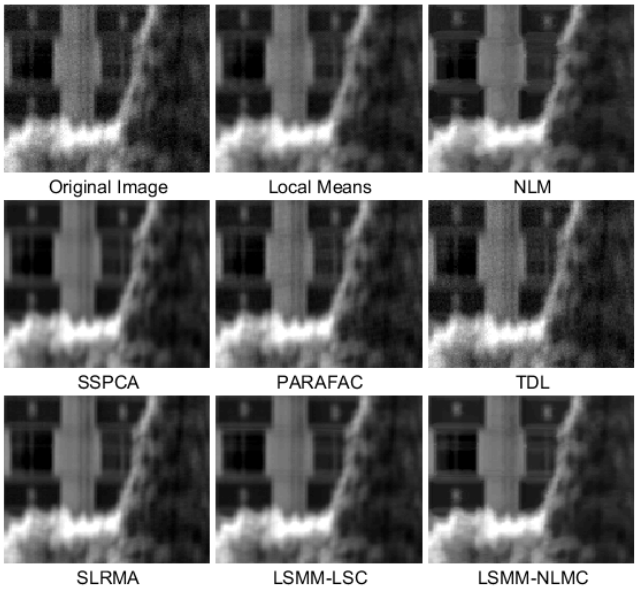
Fig. 9 shows a violin plot of MSNR values for all the spectral bands of the two SOC710-VP data sets after and before denoising. The results show that the MSNR values of the original HSI for both scenes are increased by all the denoising methods. For the distribution and quartile of MSNR values, the proposed LSMM-NLMC method has the most concentration result and shorter range of quartile than other denoising methods, which means that it adaptively removes noise for different bands. For the median of MSNR values, the proposed LSMM-NLMC method also obtains the maximum value, which means that it has the best denoising capacity than other competitors.

The visualization results of the 4th and 119th spectral bands on data set 1 and data set 2 are presented in Figs. 10 and 11, respectively. From Figs. 10 and 11, we can observe that the original HSI bands are polluted by complex noise with different levels. The local means and TDL methods cannot remove the noises completely, and the PARAFAC and NLM methods bring some structural noise in some bands. The SSPCA and LSMM-LSC methods lose the structural information seriously. The SLRMA methods give a better result, but it also loses some information compared with the proposed method. The proposed LSMM-LSC method receives a better visual quality than the other eight

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(a)



(b)

Fig. 11. Spectral band images of the SOC710-VP data set 2 before and after denoising with the eight methods. (a) 4th band. (b) 119th band.

methods, which is smooth in the regions and sharp on the edge.

The presented denoising results demonstrate that the effectiveness of the LSMM-NLMC method in HSI denoising and distinguishing the wanted signal from noise in the mixture spectral is better than of the other comparable methods. The nonlocal spatial–spectral constraint provided by the improved NLM method strengthens the recognition ability of the LSMM-NLMC method in noise detection.

F. Computational Complexity Analysis

Assuming that the size of HSI is $m1 \times m2 \times P (N = m1 \times m2)$, the computation cost of the NLMC in the proposed method is $N^2 \times f^2 \times l^2 \times P$ by fixing the size of 3-D-block as $\times f^2$ in the spatial domain. The "search window" size is fixed as l^2. The computation cost of estimating $\boldsymbol{A}$ and $\boldsymbol{S}$ is $P \times K$ and $K \times N$. Thus, the overall computational complexity of the proposed LSMM-NLSC method per iteration is $N^2 \times f^2 \times l^2 \times P + P \times K + K \times N$. But the compared methods do not perform computational complexity analysis in the original paper.

Overall the algorithm is not very computationally efficient, mainly due to the implementation uses MATLAB that tends to be slow, and also the unsupervised iterative nature of the algorithm that requires many iterations to update the endmember estimation. In the future, we could implement the proposed approach under more efficient algorithms such as C++ with a graphical processing unit (GPU) CUDA capability for boosting the computational efficiency.

VI. Conclusion

In this article, an LSMM-NLMC method for more accurate and effective denoising of mixture spectral of HSI is proposed. The LSMM-NLMC method considers three main characteristics of HSI, a large account of mixture spectra, spatial nonstationarity, and noise variance heterogeneity, to achieve more accurate and reliable denoising. Based on the LSMM with nonlocal spatial–spectral constraint, the LSMM-NLMC method is developed using the EM algorithm under the maximum-likelihood estimator (MLE) framework. The nonlocal spatial–spectral constraint consists of a set of neighboring spectral samples and their spatial–spectral self-similarity with the reference spectra. The spatial–spectral self-similarity is computed using the NLM method improved by the Mahalanobis distance in nonlocal and local regions. The proposed method is validated by the experiments on both simulated and real data sets, and experimental results show both numerically and visually the effectiveness of the LSMM-NLMC method in noise removal. Finally, the LSMM-NLMC method can more effectively differentiate wanted signals from noise and retain more edge information.

However, the proposed method has some shortcomings. Namely, the quality of denoising results is affected by the noise variances. Therefore, our future work will be focused on estimating more accurate noise variances to extend the HSI application even more.

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References

[1] D. Bannon, "Hyperspectral imaging: Cubes and slices," *Nature Photon.*, vol. 3, no. 11, pp. 627–629, 2009.

[2] R. Heylen, M. Parente, and P. Gader, "A review of nonlinear hyperspectral unmixing methods," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 1844–1868, Jun. 2014.

[3] R. Siche, R. Vejarano, V. Aredo, L. Velasquez, E. Saldaña, and R. Quevedo, "Evaluation of food quality and safety with hyperspectral imaging (HSI)," *Food Eng. Rev.*, vol. 8, no. 3, pp. 306–322, Sep. 2016.

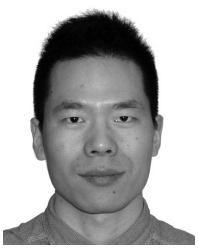
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 17:00:55 UTC from IEEE Xplore. Restrictions apply.

[4] R. Zimmermann, E. Herrmann, D. Rosa, M. Kirsch, and R. Gloaguen, "Hyperspectral imaging for structural geology: An example for more interdisciplinary data analysis in mineral exploration," in *Proc. EARSel SIG Imag. Spectroscopy Workshop*, 2017.
[5] K. Tiwari, M. Arora, and D. Singh, "An assessment of independent component analysis for detection of military targets from hyperspectral images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 13, no. 5, pp. 730–740, Oct. 2011.
[6] G. Bearman and C. Van Veen, "Hyperspectral imaging opens new perspectives in cultural heritage," *Laser Focus World*, vol. 51, no. 5, pp. 54–56, 2015.
[7] H. Chung, G. Lu, Z. Tian, D. Wang, Z. G. Chen, and B. Fei, "Superpixel-based spectral classification for the detection of head and neck cancer with hyperspectral imaging," *Proc. SPIE*, vol. 9788, Mar. 2016, Art. no. 978813.
[8] J. B. Gladden, "2002 hyperspectral analysis of hazardous waste sites on the savannah river site," Office Sci. Tech. Inf. Tech. Rep., Tech. Rep., 2003.
[9] N. Renard, S. Bourennane, and J. Blanc-Talon, "Denoising and dimensionality reduction using multilinear tools for hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 2, pp. 138–142, Apr. 2008.
[10] A. Karami, M. Yazdi, and A. Zolghadre Asli, "Noise reduction of hyperspectral images using Kernel non-negative tucker decomposition," *IEEE J. Sel. Topics Signal Process.*, vol. 5, no. 3, pp. 487–493, Jun. 2011.
[11] J. Ho and W.-L. Hwang, "Wavelet Bayesian network image denoising," *IEEE Trans. Image Process.*, vol. 22, no. 4, pp. 1277–1290, Apr. 2013.
[12] X. Zheng, Y. Yuan, and X. Lu, "Hyperspectral image denoising by fusing the selected related bands," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2596–2609, May 2019.
[13] W. Liu and J. Lee, "A 3-D atrous convolution neural network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5701–5715, Aug. 2019.
[14] Y. Yuan, X. Zheng, and X. Lu, "Spectral–spatial Kernel regularized for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3815–3832, Jul. 2015.
[15] D. Cerra, R. Müller, and P. Reinartz, "Noise reduction in hyperspectral images through spectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 109–113, Jan. 2014.
[16] H. Hotelling, "Analysis of a complex of statistical variables into principal components.," *J. Educ. Psychol.*, vol. 24, no. 7, pp. 498–520, 1933.
[17] G. Chen and S.-E. Qian, "Simultaneous dimensionality reduction and denoising of hyperspectral imagery using bivariate wavelet shrinking and principal component analysis," *Can. J. Remote Sens.*, vol. 34, no. 5, pp. 447–454, Oct. 2008.
[18] G. Chen and S.-E. Qian, "Denoising and dimensionality reduction of hyperspectral imagery using wavelet packets, neighbour shrinking and principal component analysis," *Int. J. Remote Sens.*, vol. 30, no. 18, pp. 4889–4895, Aug. 2009.
[19] G. Chen and S.-E. Qian, "Denoising of hyperspectral imagery using principal component analysis and wavelet shrinkage," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 3, pp. 973–980, Mar. 2011.
[20] J. Huang, Y. Ma, X. Mei, and F. Fan, "A hybrid spatial-spectral denoising method for infrared hyperspectral images using 2DPCA," *Infr. Phys. Technol.*, vol. 79, pp. 68–73, Nov. 2016.
[21] B. Aiazzi *et al.*, "Noise modelling and estimation of hyperspectral data from airborne imaging spectrometers," *Ann. Geophys.*, vol. 49, no. 1, pp. 1–9, 2006.
[22] A. S. Fotheringham, M. E. Charlton, and C. Brunsdon, "Geographically weighted regression: A natural evolution of the expansion method for spatial data analysis," *Environ. Plan A*, vol. 30, no. 11, pp. 1905–1927, Nov. 1998.
[23] S. Villena, M. Vega, R. Molina, and A. Katsaggelos, "A non-stationary image prior combination in super-resolution," *Digit. Signal Process.*, vol. 32, pp. 1–10, Sep. 2014.
[24] A. Buades, B. Coll, and J. M. Morel, "A review of image denoising algorithms, with a new one," *Multiscale Model. Simul.*, vol. 4, no. 2, pp. 490–530, Jan. 2005.
[25] J. Li, Q. Yuan, H. Shen, and L. Zhang, "Hyperspectral image recovery employing a multidimensional nonlocal total variation model," *Signal Process.*, vol. 111, pp. 230–248, Jun. 2015.
[26] L. Sun and J.-S. Luo, "Junk band recovery for hyperspectral image based on curvelet transform," *J. Cent. South Univ. Technol.*, vol. 18, no. 3, pp. 816–822, Jun. 2011.
[27] Q. Yuan, L. Zhang, and H. Shen, "Hyperspectral image denoising employing a spectral–spatial adaptive total variation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3660–3677, Oct. 2012.
[28] Z. Huang, S. Li, L. Fang, H. Li, and J. A. Benediktsson, "Hyperspectral image denoising with group sparse and low-rank tensor decomposition," *IEEE Access*, vol. 6, pp. 1380–1390, 2018.
[29] J. Xue, Y. Zhao, W. Liao, and J. C.-W. Chan, "Nonlocal low-rank regularized tensor decomposition for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 5174–5189, Jul. 2019.
[30] X. Bai, F. Xu, L. Zhou, Y. Xing, L. Bai, and J. Zhou, "Nonlocal similarity based nonnegative tucker decomposition for hyperspectral image denoising," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 701–712, Mar. 2018.
[31] Y. Qian, Y. Shen, M. Ye, and Q. Wang, "3-D nonlocal means filter with noise estimation for hyperspectral imagery denoising," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2012, pp. 1345–1348.
[32] L. Xu, A. Wong, F. Li, and D. A. Clausi, "Extraction of endmembers from hyperspectral images using a weighted fuzzy purified-means clustering model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 2, pp. 695–707, Feb. 2016.
[33] L. Xu, A. Wong, F. Li, and D. A. Clausi, "Intrinsic representation of hyperspectral imagery for unsupervised feature extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 2, pp. 1118–1130, Feb. 2016.
[34] J.-S. Lee, "Refined filtering of image noise using local statistics," *Comput. Graph. Image Process.*, vol. 15, no. 4, pp. 380–389, Apr. 1981.
[35] X. Liu, S. Bourennane, and C. Fossati, "Denoising of hyperspectral images using the PARAFAC model and statistical performance analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3717–3724, Oct. 2012.
[36] Y. Peng, D. Meng, Z. Xu, C. Gao, Y. Yang, and B. Zhang, "Decomposable nonlocal tensor dictionary learning for multispectral image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 2949–2956.
[37] W. Wei, L. Zhang, Y. Jiao, C. Tian, C. Wang, and Y. Zhang, "Intracluster structured low-rank matrix analysis method for hyperspectral denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 866–880, Feb. 2019.
[38] *Hyperspectral Remote Sensing Scenes*. [Online] Available:http://engineering.purdue.edu/biehl/MultiSpec/hyperspectral.html
[39] G. A. Swayze, R. N. Clark, T. V. V. King, A. Gallagher, and W. M. Calvin, "The US geological survey, digital spectral library: Version 1: 0.2 to 3.0 m," *Bull. Amer. Astron. Soc.*, vol. 25, p. 1033, Jun. 1993.



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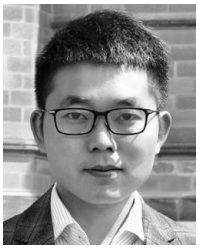
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